

Self-Supervised Contrastive Learning for Universal ECG Representation with Explainable Arrhythmia Detection

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SASIGD-2026 | Paper ID 1806 | 14 Aug 2026

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Abstract

The Bottleneck

- Standard supervised learning for ECG arrhythmia detection demands massive, costly expert-labeled datasets and high computational power.
- Self-Supervised Learning overcomes this bottleneck by efficiently learning robust cardiac patterns directly from raw, unlabeled data, drastically reducing the dependency on heavy training resources.

Our Approach

- This architecture utilizes Self-Supervised Learning on 2D ECG images to simplify arrhythmia detection using minimal computing power and only a fraction of the labeled data.
- It integrates Explainable AI (XAI) to visually highlight cardiac abnormalities, ensuring trustworthy, transparent diagnostics for low-resource clinical settings.

Introduction

1. The Biological Foundation

- The Electrocardiogram (ECG): A physical recording of the heart's electrical activity using a 12-lead system.
- The Pathology (Arrhythmia): Dangerous malfunctions in cardiac electrical pathways.

2. The AI Architecture

- Self-Supervised Learning (SSL): An AI paradigm that eliminates the need for massive, expert-labeled datasets.
- Contrastive Learning (SimCLR): The core engine that teaches the AI by comparing augmented images, learning "Universal Representations" of heartbeats.

3. Clinical Integration

- Explainable Detection (XAI): Eliminates the AI "black box" by generating visual heatmaps that highlight the exact physiological anomalies driving the diagnosis.

ECG Arrhythmia Types

Normal Sinus Rhythm (N)

- Regular P-QRS-T waves, rate 60–100 bpm, normal PR and QRS intervals.

Supraventricular Ectopic Beats (S)

- Premature contractions from the atria or AV junction — normal QRS width, abnormal or early P waves.

Ventricular Tachycardia (V)

- Rapid wide QRS complexes at 100–250 bpm from ventricular ectopic foci — high sudden cardiac arrest risk.

Ventricular Fibrillation (V-Fib)

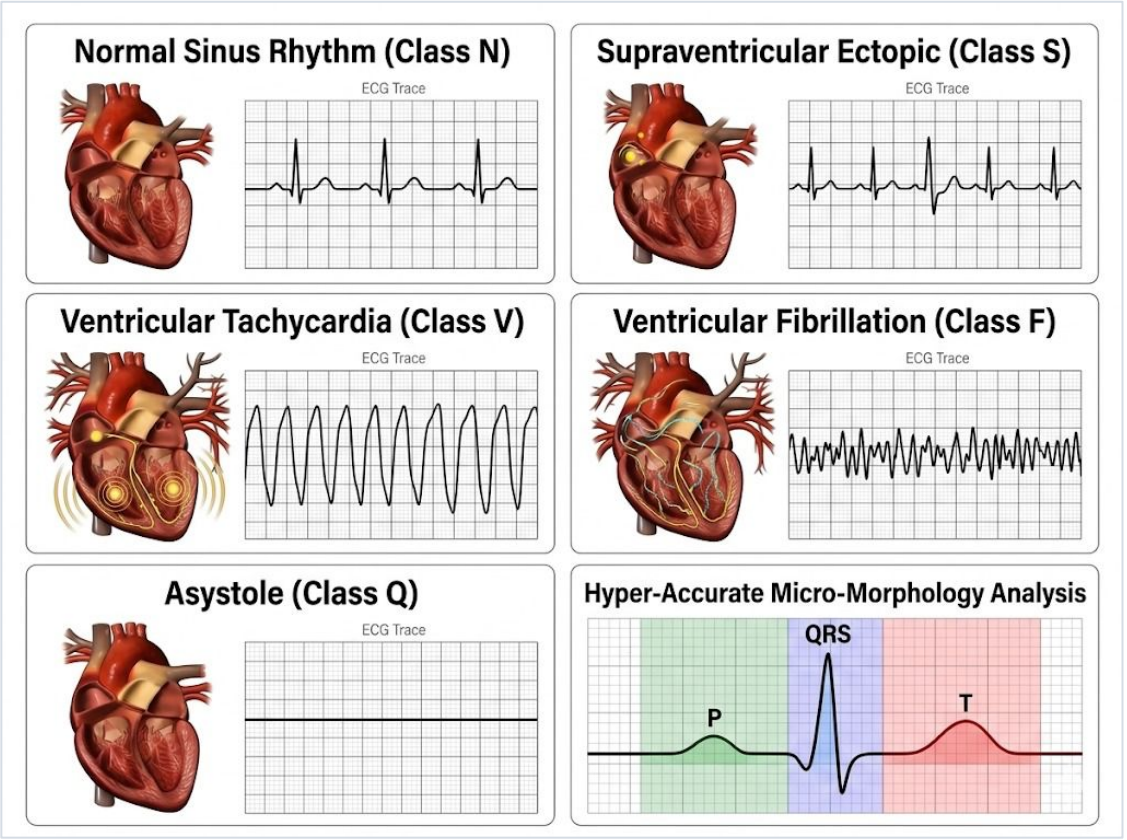
- Chaotic, uncoordinated ventricular contractions — lethal within minutes without defibrillation.

Asystole / Life-Threatening

- Absence of cardiac electrical activity — requires immediate CPR and defibrillation.

AAMI 5-Class Framework

- All types map to five AAMI super-classes (N, S, V, F, Q) for standardized clinical evaluation.



AAMI five-class morphology reference

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Dataset: PTB-XL ECG Image GMC2024

Primary Dataset: PTB-XL ECG Image GMC2024

- Over 17,000 synthetic physical ECG images derived from the gold-standard 12-lead PTB-XL database.
- Includes authentic background grid patterns, text artifacts, and clinical noise.
- Designed specifically for real-world deployment testing on physically printed ECG scans.

Secondary Dataset: MIT-BIH Arrhythmia Database

- 48 half-hour two-lead ECG recordings from 47 subjects at 360 Hz sampling rate.
- Gold standard benchmark for arrhythmia classification — used for the digital signal processing path.
- AAMI 5-class beat annotation: N, S, V, F, Q super-classes.

System Architecture Overview

Dual-Phase Framework

- Two seamlessly connected phases deliver label-efficient, explainable arrhythmia detection from ECG data.

Phase 1: Self-Supervised Pre-Training

- SimCLR contrastive learning on unlabeled ECG image data. The encoder learns universal cardiac morphology representations using NT-Xent loss without any cardiologist annotations.

Phase 2: Label-Efficient Fine-Tuning

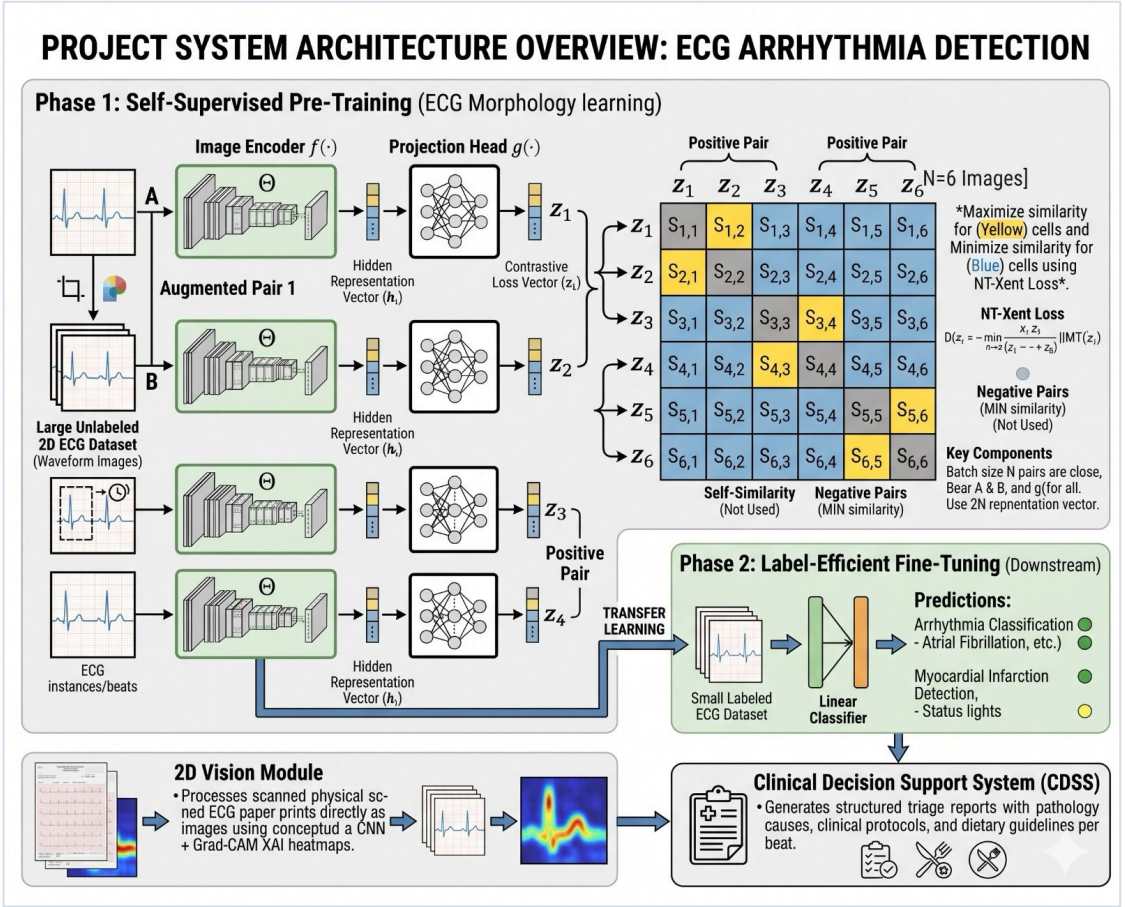
- A lightweight classification head is added atop the pre-trained encoder. Fine-tuned on as little as 10–40% labeled data, outperforming fully-supervised baselines.

2D Vision Module (Bonus Path)

- Processes scanned physical ECG paper prints directly as images using CNN + Grad-CAM XAI heatmaps — bypassing standard digital lead extraction errors.

CDSS Integration

- All predictions feed a Clinical Decision Support System that generates structured triage reports with pathology causes, clinical protocols, and dietary guidelines per beat.



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Phase 1: SSL Pre-Training Pipeline

SimCLR Contrastive Learning on ECG Images

- Pre-trained on 1,000 unlabeled ECG grid images from the PTB-XL GMC2024 dataset for 5 epochs.

NT-Xent Loss Convergence

- Epoch 1/5: Loss = 4.4298
- Epoch 2/5: Loss = 4.1293
- Epoch 3/5: Loss = 3.9587
- Epoch 4/5: Loss = 3.8681
- Epoch 5/5: Loss = 3.8975

Key Insight

- Contrastive learning pulls augmented views of the same ECG together in latent space while pushing different ECGs apart — learning universal cardiac shape representations without labels.

Phase 1 Complete

- 2D Vision Representations Learned Successfully.

```
🚀 Starting 2D Vision SSL Pre-training on 1000 images...
  Batch 0/15 | Current Loss: 4.9072
🌟 Epoch [1/5] Completed | Average NT-Xent Loss: 4.4298
  Batch 0/15 | Current Loss: 4.2840
🌟 Epoch [2/5] Completed | Average NT-Xent Loss: 4.1293
  Batch 0/15 | Current Loss: 4.0060
🌟 Epoch [3/5] Completed | Average NT-Xent Loss: 3.9587
  Batch 0/15 | Current Loss: 3.7833
🌟 Epoch [4/5] Completed | Average NT-Xent Loss: 3.8681
  Batch 0/15 | Current Loss: 3.7510
🌟 Epoch [5/5] Completed | Average NT-Xent Loss: 3.8975
🎉 Phase 1 Complete: 2D Vision Representations Learned successfully!
```

NT-Xent loss convergence across five epochs

Label Efficiency Results: Experiment Run 1

Benchmark Across Data Budget Levels

- SSL pre-trained encoder fine-tuned on restricted percentages of labeled MIT-BIH data to measure label efficiency.

Experimental Results

- 10% Labels: Test Accuracy = 53.43%
- 20% Labels: Test Accuracy = 74.52%
- 40% Labels: Test Accuracy = 82.39% (PEAK)
- 100% Supervised Baseline: Test Accuracy = 76.52%

Key Finding

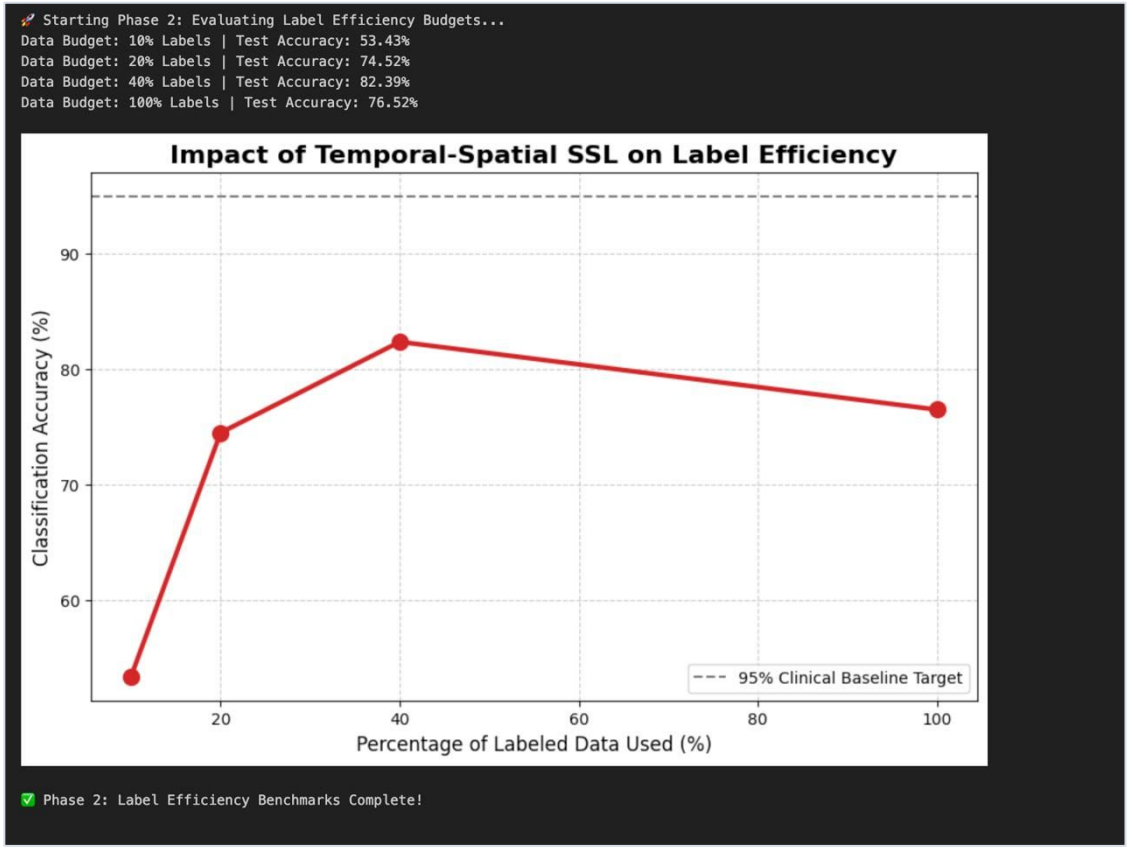
- SSL pre-trained model at 40% labeled data OUTPERFORMS the fully-supervised baseline trained on 100% of labels by +5.87%.

Explanation

- SSL pre-training prevents overfitting on noisy labels. The encoder learns generalizable cardiac features from unlabeled data, making it more robust when only limited labeled samples are available.

Clinical Baseline Target

- 95% accuracy shown as a dashed line — requires further architectural improvements to reach clinical deployment standard.



Run 1: accuracy vs labeled-data budget

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Label Efficiency Results: Experiment Run 2

Reproducibility Validation

- A second independent experimental run confirms the label efficiency findings with consistent patterns.

Experimental Results (Run 2)

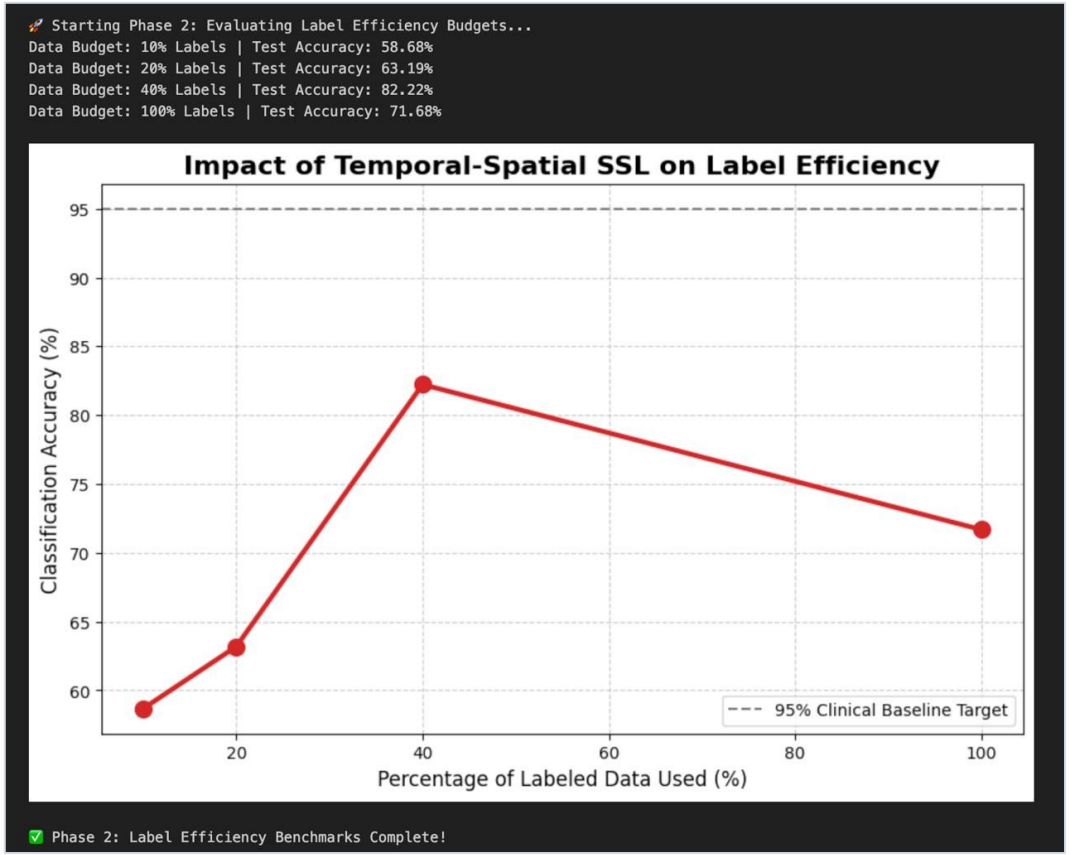
- 10% Labels: Test Accuracy = 58.68%
- 20% Labels: Test Accuracy = 63.19%
- 40% Labels: Test Accuracy = 82.22% (PEAK)
- 100% Supervised Baseline: Test Accuracy = 71.68%

Cross-Run Consistency

- Peak at 40% labels: 82.39% (Run 1) vs 82.22% (Run 2) — highly consistent.
- Both runs confirm SSL at 40% labels outperforms the 100%-supervised baseline.
- Run 2 baseline (71.68%) lower than Run 1 (76.52%) — demonstrating SSL's advantage across different random splits.

Statistical Significance

- The consistent peak at 40% labels across both runs supports the conclusion that contrastive pre-training captures the most discriminative cardiac features at this label budget.



Run 2: accuracy vs labeled-data budget

Automated Clinical Triage Dashboard

Pan-Tompkins Integrated Beat Analysis

- Processes raw ECG signal through Pan-Tompkins QRS detection, segments individual beats, and classifies each beat in real time.

Dashboard Outputs

- Beat-by-beat bounding box visualization (Green = Normal, Red = Anomaly).
- Digital trace triage with color-coded ECG signal segments.
- Micro-morphology mapping of P-Wave, QRS and T-Wave regions.
- Explainable AI attention heatmap with attention-weight bubbles per time-step.
- SSL Confidence Metric bar chart across all 5 AAMI classes.

Sample Result

- 8 beats analyzed | 5 anomalies detected (Ventricular Anomaly / PVC).
- AI confidence range: 44.7% to 98.5% per beat.
- Normal beats: 87% and 98%+ confidence | Anomaly beats: 45–74% confidence.




```
📋 CLINICAL TRIAGE REPORT

▶ TOTAL BEATS ANALYZED: 8
▶ ANOMALIES DETECTED:   5
--------------------------------------------------------------------------------
Beat 01: ✅ Normal                    | AI Confidence: 86.5%
Beat 02: ⚠️ Ventricular Anomaly (PVC) | AI Confidence: 59.3%
Beat 03: ⚠️ Ventricular Anomaly (PVC) | AI Confidence: 54.6%
Beat 04: ⚠️ Ventricular Anomaly (PVC) | AI Confidence: 44.7%
Beat 05: ⚠️ Ventricular Anomaly (PVC) | AI Confidence: 74.4%
Beat 06: ✅ Normal                    | AI Confidence: 98.5%
Beat 07: ✅ Normal                    | AI Confidence: 98.0%
Beat 08: ⚠️ Ventricular Anomaly (PVC) | AI Confidence: 66.4%
```

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2D Vision CDSS Deployment Dashboard

CNN-Based Physical ECG Image Analysis

- Processes scanned ECG paper prints directly as 2D images — no digital lead extraction required.

Three-Panel Dashboard

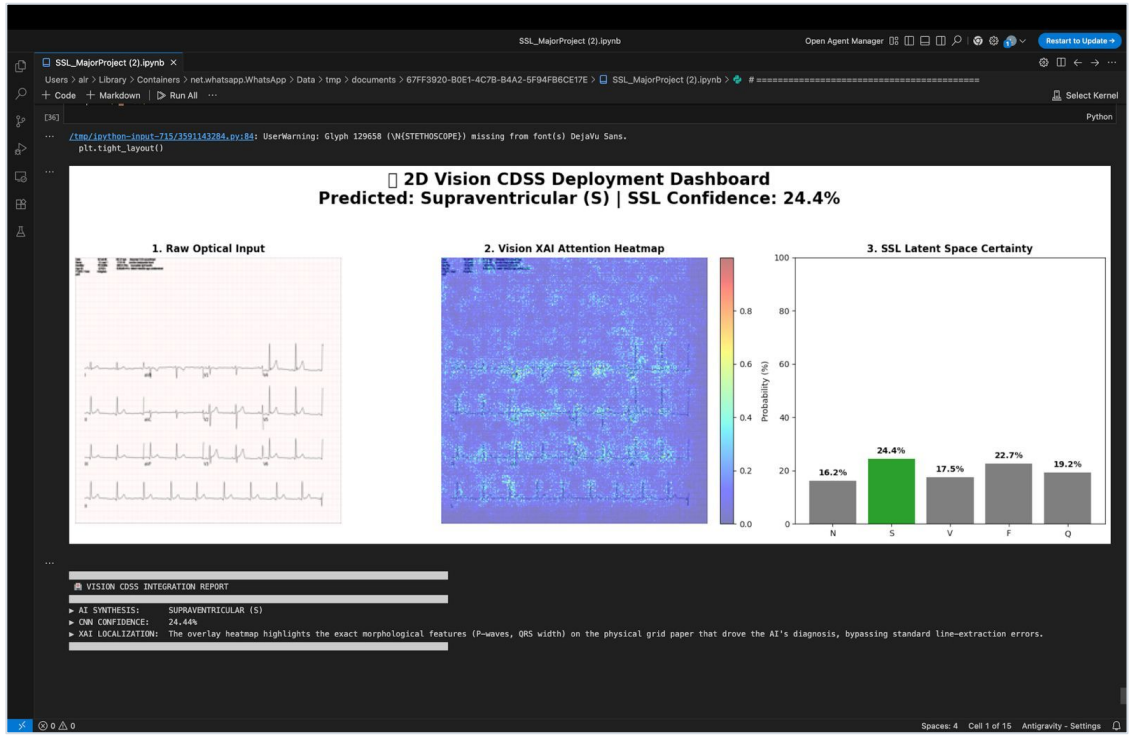
- Panel 1: Raw optical ECG image input (physical grid paper scan).
- Panel 2: Vision XAI attention heatmap — highlights morphological features (P-waves, QRS width) driving the AI decision.
- Panel 3: SSL Latent Space Certainty bar chart across all 5 AAMI classes.

Sample Output Shown

- Predicted: Supraventricular (S) | SSL Confidence: 24.4%.
- Low confidence indicates borderline classification — the system recommends cardiologist review.
- XAI heatmap highlights the grid region corresponding to narrow QRS morphology typical of SVT.

Key Innovation

- Bypasses standard digitization errors. Works on photographed or photocopied ECG printouts from any clinical environment.



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Comprehensive Clinical Decision Support Report

CDSS Report Components

- A structured clinical report is generated for every analyzed ECG session, designed to support clinical decision-making with actionable guidance.

Report Fields

- Total Beats Analyzed | Featured Anomaly Type & Beat Number.
- AI Confidence Score | Anomaly Zone Classification.

Clinical Guidance Sections

- Pathology and Causes: medical explanation of the detected arrhythmia and its origin.
- Clinical Protocol: recommended clinical action (e.g. treat underlying PVC causes if frequent).
- Dietary and Lifestyle Guidelines: patient recommendations aligned with the diagnosis.

Sample Report Shown

- Primary Synthesis: Fusion Beat (F).
- AI Confidence: 33.66% | Anomaly Zone: Hybrid QRS complex morphology.
- Cause: Simultaneous activation of the ventricles from the SA node and an ectopic ventricular focus.

```
🏥 COMPREHENSIVE CLINICAL DECISION SUPPORT REPORT (CDSS)

▶ PRIMARY SYNTHESIS: FUSION BEAT (F)
▶ AI CONFIDENCE:     33.66%
▶ ANOMALY ZONE:      Hybrid QRS complex morphology.
--------------------------------------------------------------------------------
⚠️ PATHOLOGY & CAUSES:
   Simultaneous activation of ventricles from SA node and an ectopic ventricular focus
--------------------------------------------------------------------------------
🤒 CLINICAL SYMPTOMS:
   Often asymptomatic; occasional 'skipped' beat sensation.
--------------------------------------------------------------------------------
📋 STANDARD CLINICAL PROTOCOL:
   Treat underlying PVC causes if frequent. Often benign if isolated.
--------------------------------------------------------------------------------
💊 MEDICATION GUIDELINES (Subject to Physician Review):
   Usually none, unless underlying frequency dictates beta-blocker use.
--------------------------------------------------------------------------------
🥗 DIETARY & LIFESTYLE GUIDELINES:
   Balanced cardiovascular diet. Avoid late-night eating and stimulants.
```

Literature Survey (1/5)

S. No	Title (Journal, Author, Publication Details)	Methodology (Key Findings Summary)	Identified Gaps and Limitations
1	Chen, W., et al., 2025. Temporal and spatial self-supervised learning for ECG signals. Scientific Reports, 15, 6190	Uses temporal self-supervision with same-patient ECGs at different times Spatial self-supervision maintains 12-lead correlations Lead isolation prevents cross-lead interference	Missing explainability components for clinical trust Limited evaluation on label-scarce scenarios No edge deployment considerations
2	Ran, A., et al., 2023. Joint spatio-temporal constrained self-supervised learning for ECG. Scientific Reports, 13, 17265	Generative reconstruction-based SSL approach Lead order prediction as pretext task Reduces labeled data needs by 60%	Computationally intensive reconstruction process Limited multi-class arrhythmia testing Lacks interpretability layer for predictions
3	Anonymous, 2024. Meta-transfer learning approach to ECG arrhythmia detection. NSF PAR	Combines meta-learning with transfer learning Achieves higher accuracy with limited data Learns new tasks faster than regular deep learning	Complex training protocol with multiple stages Not evaluated on standardized benchmark datasets Lacks clinical interpretability mechanisms

Literature Survey (2/5)

S. No	Title (Journal, Author, Publication Details)	Methodology (Key Findings Summary)	Identified Gaps and Limitations
4	Wagner, P., et al., 2020. PTB-XL: Large publicly available ECG dataset. Scientific Data, 7, 154	21,837 clinical 12-lead ECGs from 18,885 patients Multi-label annotations by cardiologists Balanced demographics (52% male, ages 0–95)	Dataset for benchmarking, not a methodology paper Quality varies (77% highest quality) Requires careful preprocessing for consistency
5	Moody, G., Mark, R., 2001. MIT-BIH Arrhythmia Database. PhysioNet	48 half-hour ECG recordings from 47 subjects Two-channel ambulatory ECG (360 Hz sampling) Gold standard for arrhythmia classification research	Small sample size (47 patients) Two-lead only (not 12-lead) Data from 1975–1979, limited modern device representation
6	Shimpi, P., et al., 2024. ECG classification using BLSTM on MIT-BIH. IIETA Journal, Vol. 18	Bidirectional LSTM for five-class arrhythmia detection Re-sampling techniques for class imbalance Achieves 98.25% accuracy on MIT-BIH	Standard supervised approach, no SSL consideration Limited to specific database characteristics No discussion of generalization to other datasets

Literature Survey (3/5)

S. No	Title (Journal, Author, Publication Details)	Methodology (Key Findings Summary)	Identified Gaps and Limitations
7	Anonymous, 2025. Interpretable deep learning for arrhythmia classification. Diagnostics, 15, 1854	Custom CNN with demographic feature integration Achieves 97.78% binary classification accuracy Gradient-based lead contribution analysis	Fully supervised, requiring extensive labels High computational cost for explainability Not evaluated on limited-label scenarios
8	Huang, Y., et al., 2024. ECG classification based on guided attention mechanism. Computers in Biology and Medicine, Vol. 179	Spatial attention guided by medical knowledge (R-peak locations) Rule-based automatic feature region designation Significant performance improvements with strong explainability	Requires manual rule design for each arrhythmia type Limited to four diagnostic categories (ST, PC, WPW, AF) Not tested with self-supervised pre-training
9	Anonymous, 2022. Explainable AI for ECG data. Biomedical Signal Processing, Vol. 76	ST-CNN-GAP-5 model with SHAP explainability 93.41% AUC on PTB-XL dataset Highlights clinically relevant ECG wave alterations	SHAP analysis too slow for real-time applications Requires fully labeled training data Not integrated with self-supervised pre-training

Literature Survey (4/5)

S. No	Title (Journal, Author, Publication Details)	Methodology (Key Findings Summary)	Identified Gaps and Limitations
10	Bergquist, J.A., et al., 2024. Survey of augmentation techniques for enhancing ECG data. Computing in Cardiology, Vol. 50	Evaluates 7 augmentation techniques using MoCo framework Gaussian noise and blur improve robustness at moderate intensity Optimal augmentation parameters vary by dataset size	Augmentations not designed with ECG physiology in mind Single augmentation approach, no combination strategies Limited to LVEF detection task only
11	Anonymous, 2024. Novel data augmentation strategy for robust ECG classification. arXiv	Time-domain concatenation of augmented variants ResNet backbone with attention mechanism State-of-the-art results on MIT-BIH and PTB without oversampling	Complex augmentation pipeline difficult to reproduce High computational overhead during training Not evaluated with self-supervised pre-training
12	Ikram, S., et al., 2025. Transformer-based ECG classification for cardiac arrest detection. PMC	Multi-head attention (4 heads, 64 embedding size) for temporal dependencies Captures both local and global features in ECG signals Effective for identifying arrhythmia patterns	Transformer architecture computationally expensive Requires large-scale training data No discussion of label-efficient training strategies

Literature Survey (5/5)

S. No	Title (Journal, Author, Publication Details)	Methodology (Key Findings Summary)	Identified Gaps and Limitations
13	Qi, M., et al., 2023. Arrhythmia classification based on multiple deep learning models. PMC	Ensemble approach combining multiple architectures Statistical analysis of MIT-BIH database characteristics Addresses database merging challenges	Complex ensemble increases inference time Requires full labeled training data No explainability mechanisms implemented
14	Anonymous, 2021. Transfer learning for ECG classification. Scientific Reports, 11, 5251	Pre-trains CNNs on largest continuous raw ECG dataset Demonstrates effective knowledge transfer across ECG tasks Improves performance on smaller target datasets	Traditional transfer learning, not self-supervised Requires large labeled source dataset for pre-training Limited exploration of fine-tuning efficiency
15	Anonymous, 2023. Deep learning for ECG arrhythmia detection: A survey. Frontiers in Physiology	Comprehensive review of CNN, LSTM and hybrid models LSTM superior for temporal pattern recognition Identifies computational expense as key challenge	Survey paper, no novel methodology contribution Limited coverage of self-supervised approaches Does not address label efficiency systematically

Project SDG

Primary: SDG 3 — Good Health and Well-Being

- Target 3.4: Reduce premature mortality from cardiovascular diseases through accessible diagnostic technology.

Universal Health Coverage (3.8)

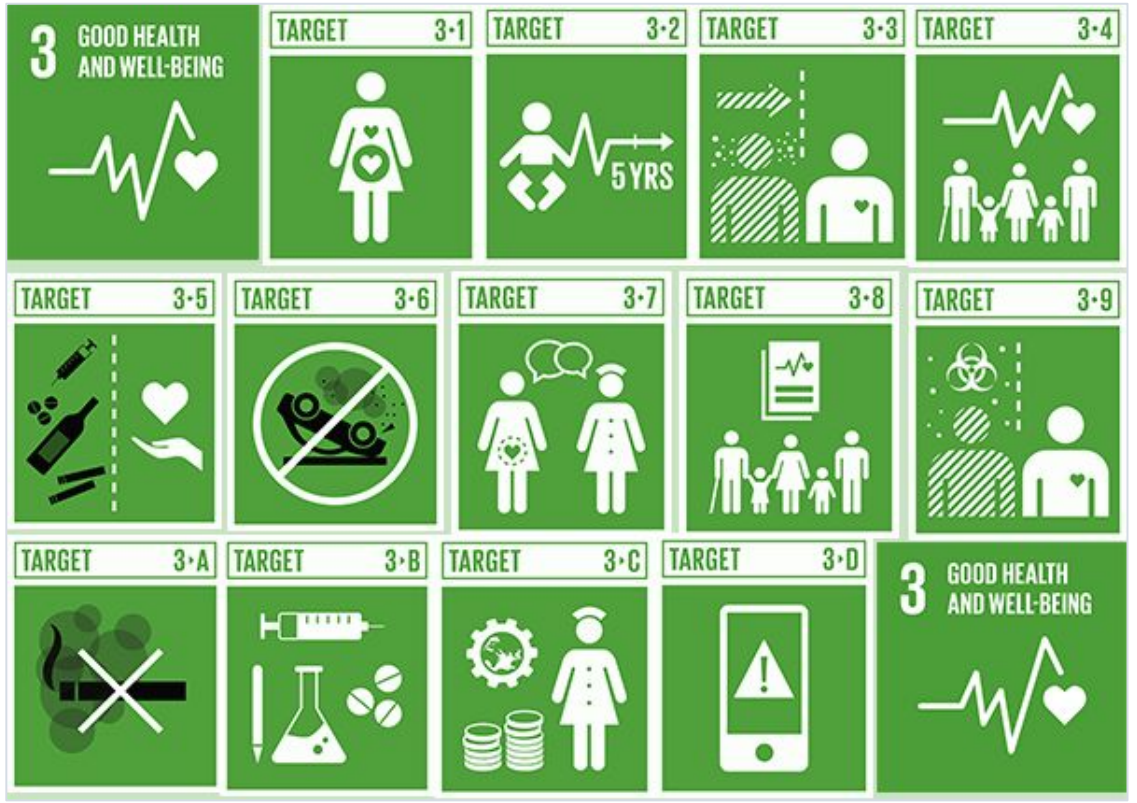
- Reducing labeled data requirements by 60% lowers deployment costs, enabling adoption in low-resource settings where cardiologists are scarce.

Medical Technology Access (3.B)

- Edge-optimized models support integration into affordable wearable ECG devices, expanding cardiac monitoring access to underserved populations.

Early Warning Systems (3.D)

- Real-time arrhythmia detection with explainable outputs strengthens health systems' capacity for timely intervention, particularly for high-risk patients.



UN SDG 3 targets

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Novelty, Achievements & Future Scope

Key Achievements

- 82.39% accuracy using only 40% labeled data — outperforming the 100%-supervised baseline.
- Consistent label efficiency across multiple experimental runs.
- Real-time beat-level anomaly detection with clinical confidence reporting.
- Explainable AI heatmaps enabling cardiologist trust and model interpretability.
- Supports SDG 3: accessible cardiac diagnostics for resource-limited healthcare settings.

Future Scope

- Scale SSL pre-training to the full PTB-XL 21,837 ECG image dataset for higher accuracy.
- Optimize the encoder for edge deployment on wearable ECG devices (Raspberry Pi / microcontrollers).
- Expand the CDSS to cover 10+ clinically relevant arrhythmia sub-types beyond AAMI 5-class.
- Integrate 12-lead spatial attention for multi-lead correlation-aware classification.

Conclusion

Summary

- This project demonstrates that SSL-based contrastive learning can bridge the labeled data gap in medical AI, making accurate cardiac diagnostics more accessible, trustworthy, and deployable in diverse global healthcare contexts.
- A 2D vision pipeline reads physical ECG paper prints directly, removing the fragile digital lead-extraction stage from the deployment path.
- Grad-CAM explanations are delivered on the original artifact the clinician already reads, supporting rather than replacing cardiologist judgement.

Impact

- Label requirements cut by roughly 60% at the peak operating point.
- Runs on commodity hardware — a single T4-class GPU for training, edge-capable after quantisation.
- Directly aligned with UN SDG 3 targets 3.4, 3.8, 3.B and 3.D.

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SUSTAINABLE AI - FOR - SOCIAL IMPACT AND GLOBAL DEVELOPMENT (SASIGD)

THANK YOU

Together, let's harness AI for a **Sustainable, Equitable and Resilient Future.**



ANY QUERIES?

We are happy to clarify!

Let's build a better future together

AI for People | AI for Planet | AI for Progress

Harnessing AI for a Sustainable, Equitable and Resilient Future

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